

JERINA G

Senior Frontend Developer | Web Applications

+91 75502 37615 | inboxofjerina@gmail.com

PROFESSIONAL SUMMARY

Front-End Engineer with **6 years** building complex, production-grade React applications for global enterprise teams at Caterpillar. Delivered a **Unified Workplace Management App adopted by 1,000+ users** across 15 global facilities, from requirements gathering with stakeholders to deployment, release management, and post-launch support. Contributed to the development of an **Enterprise Chatbot** using Bot Framework **reducing tier-1 service desk volume**, and deflecting an estimated 20-30% of routine requests from live agents. Brings depth beyond the frontend through SQL, Power BI, .NET/C# backend integration, Microsoft Azure and REST API collaboration. Actively contributes to team growth through code reviews and mentoring junior developers. Explores emerging technologies including AI-augmented development to enhance existing enterprise applications.

TECHNICAL SKILLS

Core Front-End	React.js, React Hooks, JavaScript (ES6+), HTML5, CSS3, Sass/SCSS, TypeScript, Responsive & Cross-browser Design
State & Forms	Redux, Context API, Controlled-form Patterns, Session State Management
UI	Semantic HTML, Material UI, Bootstrap, Reusable Component Libraries
Testing	Unit, Integration and End-to-End Testing, code quality, performance, and deployment confidence
Backend & Cloud	REST API Integration, .NET/C# API Contracts, Microsoft Azure, SQL & Database Management
Build & CI/CD	Azure DevOps, Webpack, GitHub, Release Management, CI/CD
Mobile	React Native, Expo CLI, Android Studio Emulators
AI & NLP	GitHub Copilot, OpenAI API, Microsoft Cognitive Services, LUIS (NLP), critical evaluation of AI-generated code
Ways of Working	Agile / Scrum, Sprint Planning, Cross-functional Collaboration, Mentoring
Data & Reporting	Power BI, SQL, Utilization Dashboards, Operational Reporting

WORK EXPERIENCE

Front-End Developer / IT Analyst II – Web Applications

Jul 2020 - Present

Caterpillar India Engineering Solutions Pvt. Ltd., Bengaluru

- **Owned end-to-end delivery** of complex front-end features across multiple views, components, and applications, translating ambiguous business requirements into maintainable React.js solutions and reducing rework cycles by an estimated **20-25%** through early design alignment with stakeholders.
- **Architected and led the Unified Workplace Management App** (desk, cab, and resource bookings) serving **1,000+ users across global facilities**, built entirely in-house and **saving outsourced development costs**. Automated booking workflows that previously required **~20 manual hours/week** of coordinator effort, with real-time availability and utilization tracking.
- **Built reusable React component libraries** (Material UI, Bootstrap), cutting front-end development time for new features by an estimated **30-40%** through shared patterns and eliminating repeated component-level work across all application surfaces.
- **Aligned REST API contracts with backend engineers** (.NET/C#), reducing integration defects and shortening API-related debug cycles by an estimated **2-3 days per sprint**.
- **Maintained Azure DevOps for builds and releases**, reducing manual release coordination effort by an estimated **4-6 hours per release cycle** and improving deployment predictability.
- **Designed SQL schemas and Power BI dashboards** surfacing booking trends and utilization metrics, replacing ad-hoc spreadsheet reporting that consumed **~5 hours/week** of operational staff time and enabling data-driven facility planning.
- **Championed code quality and testing discipline** across unit, integration, and end-to-end layers, improving deployment confidence and reducing production defects on critical workflows.
- **Mentored junior developers** through structured code reviews and technical guidance; produced technical documentation covering component design decisions, API integration points, and runbook-style operational notes, reducing onboarding ramp time and raising overall team code quality.

- **Applied security-conscious development practices** aligned to Secure SDLC principles, including input validation, session and access control in the booking and admin portal, and regular dependency audits to mitigate front-end security risks.
- **Proficient with AI-assisted development tools** including GitHub Copilot and Claude, applying them to accelerate development while critically evaluating AI-generated code for correctness, security, and maintainability before committing.

Machine Learning Data Associate
Amazon India, Chennai

Jun 2018 - Jun 2020

- **Evaluated Alexa-enabled devices** for voice interaction quality and intent-recognition accuracy across diverse real-world usage scenarios, contributing to measurable model performance improvements for the India market.
- **Validated Goal Success Rate (GSR) metrics** through structured error analysis, producing prioritised findings that informed model fixes and drove iterative accuracy improvements across evaluated intent categories.
- **Designed and maintained test case documentation and sampling frameworks** across intent categories, language variations, and device configurations, improving coverage consistency and reducing duplicate test effort across the team.

KEY PROJECTS

Unified Desk & Resource Booking Application - Caterpillar

- **Delivered the full application end-to-end** from requirements through production rollout across multiple global facilities, by keeping development entirely in-house, and serving **1,000+ users** daily.
- **Built an SVG-based interactive floor plan** with live desk availability, enabling one-click seat selection and eliminating manual desk assignment, reducing front-desk coordination time by an estimated **15+ hours/week**.
- **Engineered booking conflict logic** handling concurrent reservations across users and time zones, reducing double-booking incidents to near zero after launch.
- **Delivered an admin portal** with desk-blocking, user/team management, configurable notifications, and a cab-booking export module, eliminating manual vendor coordination and saving an estimated **8-10 hours/week** of administrative effort.

Known Error Database Application - Caterpillar

- **Architected a knowledge-management app** in React.js for service desk teams to search and retrieve IT incident resolutions, cutting average resolution lookup time by an estimated **60%** vs. prior manual documentation search, and measurably reducing repeat escalations.
- **Built a dynamic search and filter interface** (keyword, category, date range, resolution type) backed by a SQL database designed end-to-end, enabling agents to resolve incidents faster and reducing average ticket handling time.

Enterprise Conversational Chatbot & AI Integration - Caterpillar

- **Led in-house development of a production enterprise chatbot** enabling employees to raise IT tickets and ask HR policy questions in natural language, reducing tier-1 service desk volume, and deflecting an estimated **20–30%** of routine requests from live agents.
- **Built multi-turn conversational flows with context retention** and integrated LUIS for intent recognition, reducing failed chatbot interactions and improving first-attempt resolution rate.
- **Integrated OpenAI and Microsoft Cognitive Services** into the desk-booking app for natural-language reservations, reducing the booking process from **5 manual steps to a single natural-language input**.

React Native Mobile Application - Caterpillar

- **Developed a mobile proof-of-concept** with React Native and Expo, validating cross-platform UI compatibility across device sizes using Android Studio emulators and establishing a viable mobile extension path for the desk-booking platform at **zero additional infrastructure cost**.

EDUCATION

Bachelor of Engineering – Electrical & Electronics
St. Joseph's Institute of Technology, Chennai | 83%

2014 – 2018

LANGUAGES

English (Professional Fluency) | Tamil (Native)